Optimal Light Spectra Under Regulatory Constraints: How Efficiency Standards Drive Lamps to Weird Spectral States

Generated by `photopic` — a Modula-3/Scheme spectral optimizer

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Abstract

Government regulations on lamp efficiency (lumens per watt) and color quality (Color Rendering Index) define a constrained optimization problem whose solutions are not the smooth, broad spectra of incandescent bulbs but narrow-band, spiky spectral power distributions. We use `photopic`, a hybrid Modula-3/Scheme numerical tool, to explore the Pareto frontier of this trade-off. As the efficacy requirement increases, the optimal spectrum converges to approximately three narrow peaks near 450, 540, and 610 nm — Thornton's "prime color" wavelengths — while aggressively suppressing power at all other wavelengths, including the red region critical for color fidelity. The results show that regulatory pressure, when pushed to its logical extreme, drives light sources toward spectra that are visually "weird": they satisfy the letter of color rendering standards while producing light that many people find unnatural.

1 Introduction

An incandescent lamp is a near-perfect blackbody radiator. Its spectrum is broad and smooth, rendering colors with perfect fidelity (CRI = 100). But most of its radiation falls in the infrared, giving it a luminous efficacy of only ∼15 lm/W. Government regulations worldwide — EU Ecodesign, US EISA/DOE, California Title 24 — have mandated minimum efficacy standards that incandescent lamps cannot meet, effectively banning them.

The replacement technologies (fluorescent, LED) achieve high efficacy by concentrating their emission in the visible spectrum. But *where* in the visible spectrum? The CIE photopic luminous efficiency function $V(\lambda)$ peaks sharply at 555 nm (yellow-green). A monochromatic source at 555 nm would achieve the theoretical maximum of 683 lm/W — but it would render no colors at all.

The question becomes: **What spectral power distribution maximizes luminous efficacy while maintaining acceptable color rendering?**

This is a constrained optimization problem. The constraints come from regulations:

- **CRI $\geq$ 80** (Energy Star, EU); CRI $\geq$ 90 (California Title 24 for some applications)
- **CCT within ±50K** of the nominal color temperature (e.g., 2700K for "warm white")
- **Duv $\leq$ 0.012**: chromaticity tolerance (distance from the blackbody locus)

We explore what happens as you "turn up the knob" on the efficacy requirement. The answer is surprising: the optimal spectra become increasingly bizarre.

2 Background

2.1 Luminous Efficacy

Luminous efficacy of radiation (LER) is the ratio of luminous flux to radiant power:

$$\eta = K_m \frac{\int_0^\infty S(\lambda)\, V(\lambda)\, d\lambda}{\int_0^\infty S(\lambda)\, d\lambda} \tag{1}$$

where $S(\lambda)$ is the spectral power distribution, $V(\lambda)$ is the CIE 1931 photopic luminous efficiency function, and $K_m = 683$ lm/W is the maximum luminous efficacy at 555 nm.

For a 2700K blackbody, the total LER is only ∼15 lm/W because most radiation is in the infrared. If we restrict to the visible band (380–770 nm), the visible-only LER is ∼88 lm/W.

2.2 Color Rendering Index (CRI)

The CRI quantifies how faithfully a light source renders colors compared to a reference illuminant (a blackbody at the same correlated color temperature). The calculation, following CIE 13.3-1995 and the colorimetric formulas in Foley and van Dam [5], involves:

1. Normalize the test spectrum to $Y = 100$
2. Find the reference blackbody at the correlated color temperature (CCT) via search in CIE 1960 MacAdam (u, v) space
3. For each of 14 Test Color Samples (TCS), compute the color shift ΔE^*_{UVW} using von Kries chromatic adaptation
4. $R_i = 100 - 4.6\, \Delta E^*_{UVW}$ for each sample; $\mathrm{CRI}(R_a) = \frac{1}{8} \sum_{i=1}^{8} R_i$

The key subtlety: only TCS 1–8 contribute to R_a. TCS 9 (saturated red) is *not* part of the official CRI score. This creates a loophole that the optimizer exploits.

2.3 Thornton's Prime Colors

In 1971, W. A. Thornton of Westinghouse showed that the most chromatically effective wavelengths for white light are three narrow bands near **450, 540, and 610 nm** — the "prime colors" [1]. Conversely, wavelengths near 500 nm and 580 nm are "anti-prime" — they contribute power without adding chromaticity discrimination. Thornton was named US National "Inventor of the Year" in 1979 for his prime-color fluorescent lamp patents [4].

The optimization results below independently rediscover Thornton's prime colors from first principles.

3 The `photopic` Tool

`photopic` is a hybrid application: performance-critical spectral calculations (Planck's law, CIE color matching functions, numerical integration) are implemented in compiled Modula-3 [11, 12] libraries, while the optimization driver, CRI calculation, and plotting are written in Scheme using MScheme [13], a Scheme interpreter embedded in the Modula-3 runtime. MScheme's `sstubgen` tool automatically generates Scheme bindings from Modula-3 interfaces, allowing the Scheme code

to manipulate compiled M3 objects (spectral arrays, integration routines, color-matching data) directly without marshalling overhead. This architecture provides the rapid prototyping benefits of a scripting language for the optimization logic while retaining compiled-language performance for the inner numerical loops.

3.1 Architecture

Component	Role
`Blackbody.m3`	Planck radiance $B_\nu(T, \nu)$
`CieSpectrum.m3`	Photopic $V(\lambda)$, conversion to lumens
`CieXyz.m3`	CIE 1931 $\bar{x}, \bar{y}, \bar{z}$ color matching
`Tcs.m3`	14 Test Color Sample reflectance spectra
`TempUv.m3`	Blackbody locus in (u, v) for CCT search
`ParametricSpectrum.m3`	Piecewise-linear log-amplitude parameterization
`COBYLA_M3.m3`	Constrained optimization (Powell's COBYLA)
`NR4p4.m3`	Romberg integration (midpoint rule)
`Bracket.m3`	Brent's method for CCT root-finding
`photopic.scm`	Optimization driver, CRI calc, plotting

3.2 Spectrum Parameterization

The optimizer works in log-amplitude space. A test spectrum is represented as:

$$S(\lambda) = \exp\big(f(\lambda)\big) \cdot B_\lambda(T_0, \lambda) \tag{2}$$

where $B_\lambda(T_0, \lambda)$ is the baseline blackbody at the target CCT, and $f(\lambda)$ is a piecewise-linear function defined by n parameters (control points evenly spaced across 380–770 nm).

The resolution increases through *hierarchical subdivision*: $n = 2 \to 3 \to 5 \to 9 \to 17 \to 33 \to 65 \to 129$. At each level, COBYLA [6] minimizes the negative efficacy subject to CRI, CCT, and chromaticity constraints. The solution from level n is interpolated to initialize level $2n - 1$.

3.3 Objective Function

The optimization target vector passed to COBYLA is:

$$\mathbf{t} = \begin{pmatrix} -\eta \\ R_a - R_a^{\min} \\ R_9 - R_9^{\min} \\ R_{\min} - (R_a^{\min} - 10) \\ \mathrm{CCT} - (T_0 - 50) \\ (T_0 + 50) - \mathrm{CCT} \\ 1000\,(0.012 - D_{uv}) \end{pmatrix} \tag{3}$$

where η is the LER, and the remaining components are inequality constraints (≥ 0 required).

4 Results

4.1 Blackbody vs. Optimized Spectrum

Figure 1 shows the dramatic difference between a 2700K blackbody and the maximum-efficacy spectrum with CRI ≥ 60.

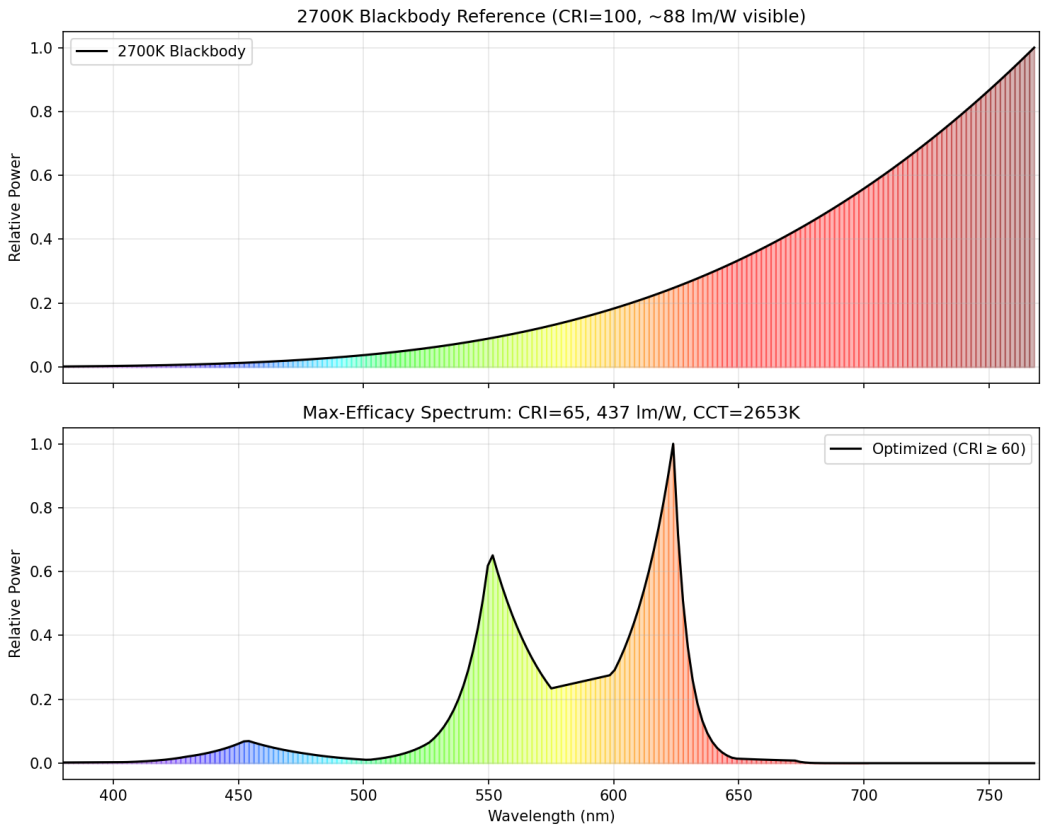

Figure 1: Top: 2700K blackbody reference spectrum (CRI = 100, ∼88 lm/W visible). Bottom: maximum-efficacy spectrum achieving CRI = 65 at 437 lm/W. The optimizer concentrates power into three peaks near Thornton's prime-color wavelengths while suppressing power below 500 nm and above 660 nm.

The optimized spectrum achieves ∼5× higher efficacy than the blackbody by eliminating "wasted" power at wavelengths where the eye is insensitive or where chromaticity is redundant. The three peaks correspond to Thornton's prime colors:

- **∼450 nm** (blue): establishes the short-wavelength chromaticity anchor
- **∼540 nm** (green): near the peak of $V(\lambda)$ — provides most of the lumens
- **∼610 nm** (orange-red): establishes warm color temperature

4.2 Convergence: The Spectrum Gets "Weirder" with More Freedom

Figure 2 shows how the optimized spectrum evolves as the number of control parameters increases from 2 to 17.

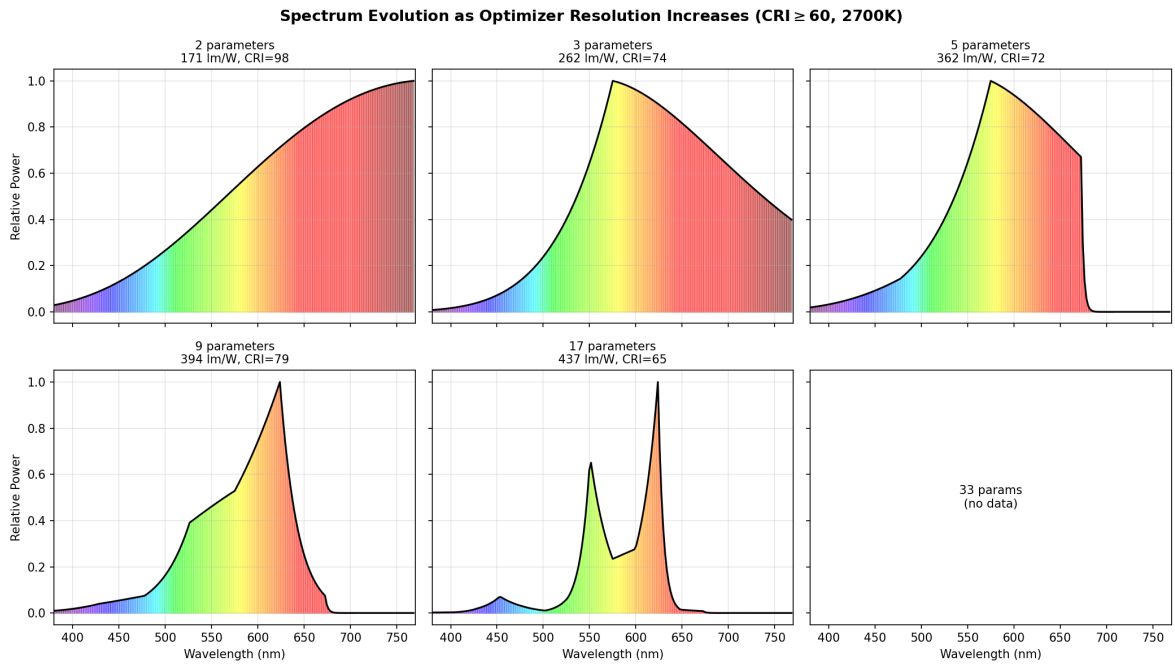

Figure 2: Evolution of the optimal CRI $\geq$ 60 spectrum at 2700K as the optimizer gains more spectral resolution. With 2 parameters, the spectrum is nearly a tilted blackbody (171 lm/W). As parameters increase, the optimizer carves out deep nulls between peaks, reaching 437 lm/W at 17 parameters. The three-peak structure emerges clearly by 9 parameters.

The progression is illuminating:

- **2 parameters** (171 lm/W): Can only tilt the spectrum. Shifts power toward green. Still smooth, CRI = 98.
- **3 parameters** (262 lm/W): First sign of spectral sculpting. Blue and deep red suppressed.
- **5 parameters** (362 lm/W): Three-peak structure begins to emerge. Deep null around 500 nm.
- **9 parameters** (394 lm/W): Clear three peaks with valleys between. CRI drops to 79.
- **17 parameters** (437 lm/W): Fully developed spiky spectrum. Deep nulls, sharp peaks. CRI just satisfies the $\geq$60 floor.

The optimizer, given enough degrees of freedom, always converges to a three-peak solution. **This is Thornton's prime-color theorem, rediscovered numerically.**

4.3 The R9 Problem

The CRI score R_a averages only TCS 1–8, which are relatively unsaturated. TCS 9 (saturated red) is reported separately. When the optimizer is free to set R_9 as low as -100, it exploits this: the CRI $\geq$ 60 optimized spectrum at 17 parameters has $R_9 = -28$.

This means: under this "optimal" light, saturated reds look dramatically wrong — less red than under a reference illuminant. A person wearing a red dress would look dull; red paint would look brownish. But the CRI score says the light is acceptable.

This is the "weird final state" that regulations drive toward: a lamp that passes the official color test while rendering an important class of colors — reds — poorly.

4.4 Effect of CRI Constraints

Figure 3 shows how the optimal spectrum changes as the CRI floor increases.

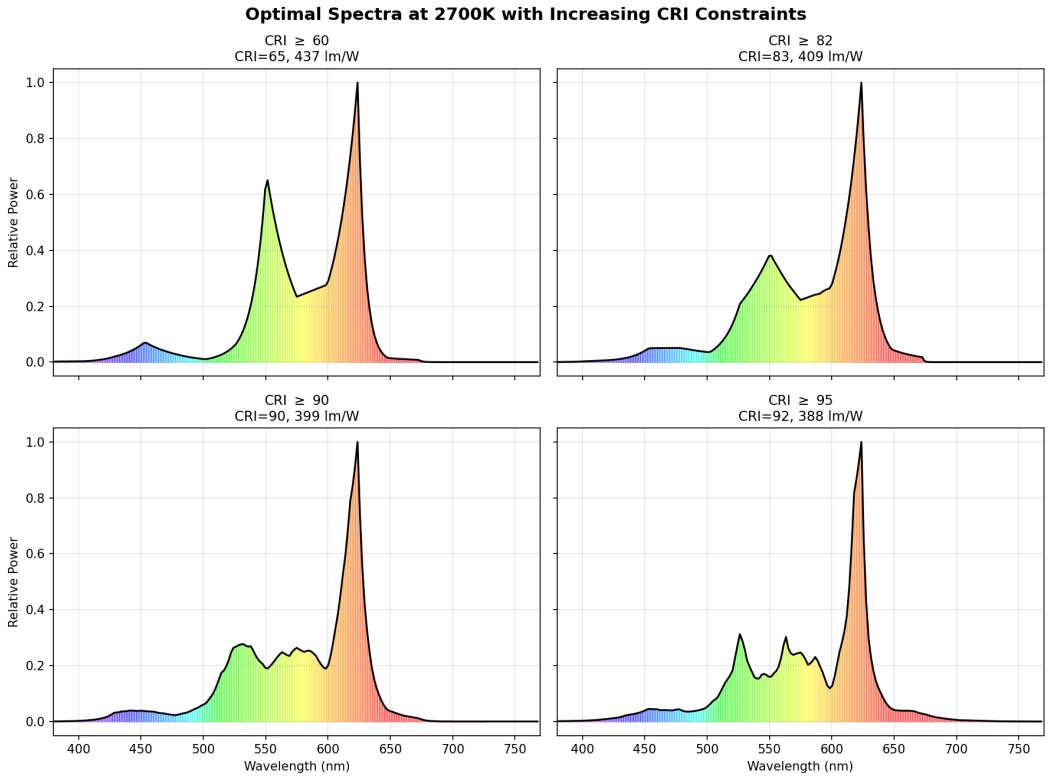

Figure 3: Optimal spectra at 2700K for different CRI constraints. As CRI requirements increase, the spectrum must broaden to render more colors faithfully, reducing the maximum achievable efficacy. The three-peak structure persists but the peaks must widen and the valleys cannot be as deep.

Table 1: Efficacy–CRI trade-off at 2700K CCT, unconstrained R_9. Results at 129 spectral parameters (17 for CRI 60).

Min CRI	Achieved CRI	Efficacy (lm/W)	R_9	Worst R_i	Character
—	100	88	100	100	2700K blackbody (visible only)
60	65	437	−28	50	Three sharp peaks, deep valleys
70	73	428	−7	60	Three peaks, moderate valleys
80	83	412	+21	70	EU minimum: R9 still poor
82	83	409	+29	72	Three peaks, shallower valleys
85	88	402	+33	75	Transitional region
90	90	399	+45	80	Broader peaks, less suppression
95	92	388	+58	86	Wide peaks, modest valleys
98	93	358	+75	89	Approaching continuous

The trade-off is dramatic: going from CRI 60 to CRI 98 costs roughly 80 lm/W of achievable efficacy (437 → 358 lm/W at 129 spectral parameters), and the key variable is R_9 — the ability to render saturated reds. At CRI 60, R_9 is deeply negative (−28), meaning reds appear wrong; at CRI 98, $R_9 = 75$ and red rendering is acceptable.

The EU regulatory minimum — CRI ≥ 80 at ∼412 lm/W — achieves high efficiency but with $R_9 = 21$, meaning reds are rendered at barely one-fifth of reference fidelity. This is the "weird final state" that market and regulatory forces converge on. California's JA8 standard, which requires $R_9 \geq 50$, pushes manufacturers above CRI 90, where the achievable efficacy drops to ∼399 lm/W but red rendering reaches $R_9 = 45$.

4.5 The Pareto Frontier: Overlay of All Solutions

Figure 4 overlays all optimized spectra on a single plot, together with the blackbody reference.

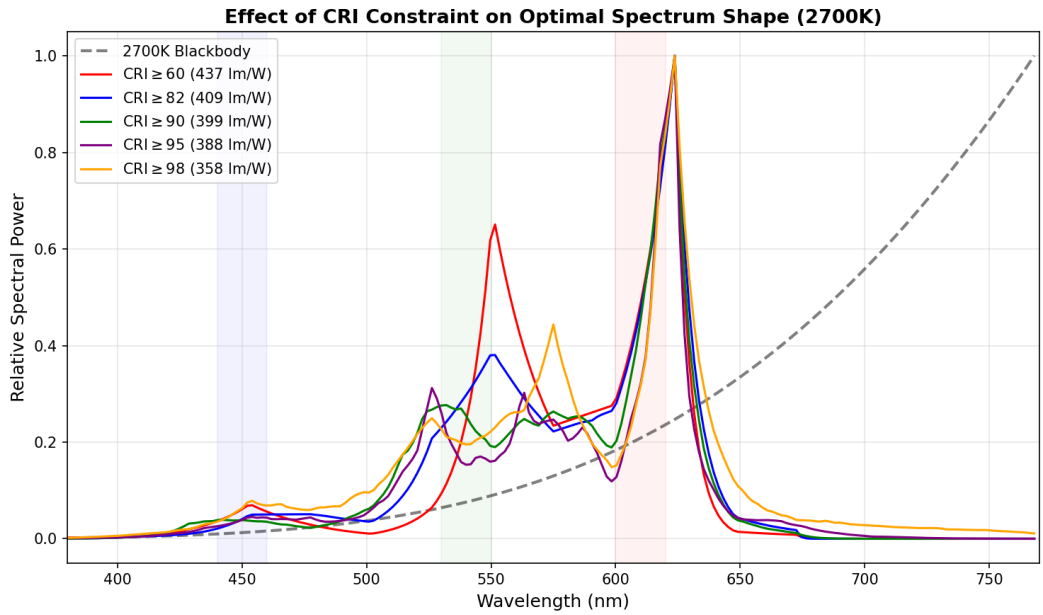

Figure 4: Overlay of optimal spectra at 2700K for CRI constraints from 60 to 98. All spectra share the same three-peak topology (blue ∼450 nm, green ∼540 nm, red ∼610 nm) but differ in peak width and valley depth. The dashed curve is the 2700K blackbody. The shaded bands mark Thornton's prime-color wavelength regions.

4.6 The Efficacy–CRI Trade-off Curve

Figure 5 plots the maximum achievable efficacy as a function of the CRI floor.

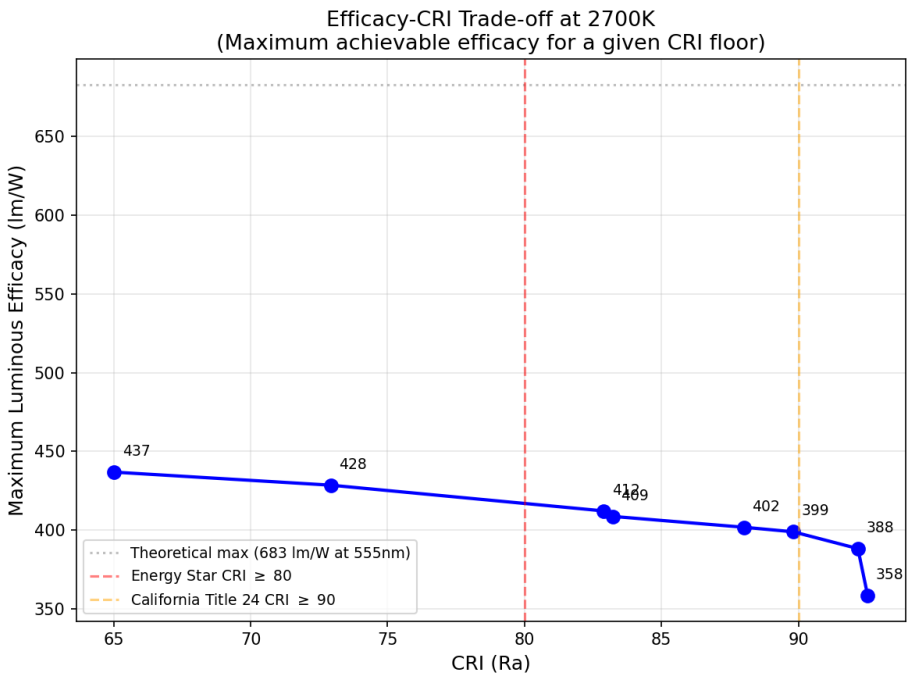

Figure 5: Pareto frontier: maximum luminous efficacy vs. minimum CRI at 2700K. Vertical dashed lines show the Energy Star (CRI $\geq$ 80) and California Title 24 (CRI $\geq$ 90) thresholds. The theoretical maximum at 555 nm is 683 lm/W. The curve shows rapidly diminishing returns: moving from CRI 82 to CRI 98 costs about 80 lm/W.

4.7 R_9 vs. Efficacy: The Red Rendering Penalty

Figure 6 shows the key result: at the EU minimum CRI $\geq$ 80, the optimal spectrum achieves 412 lm/W theoretical luminous efficacy of radiation (LER) but with $R_9 = 21$. Saturated reds are rendered at only one-fifth of their reference fidelity. To achieve California JA8's $R_9 \geq 50$ requirement, CRI must be pushed above 90, reducing achievable efficacy to $\sim$399 lm/W.

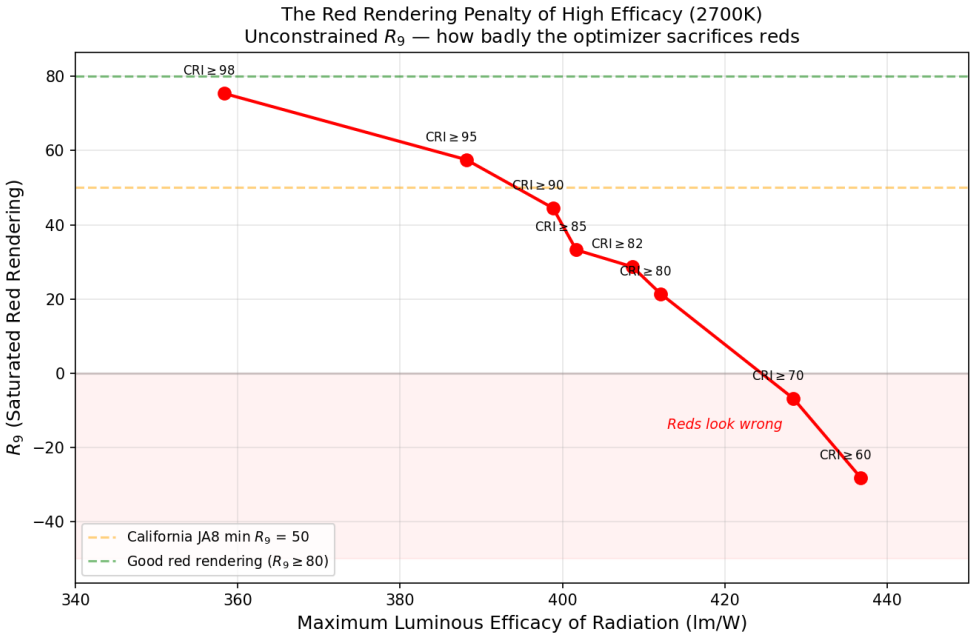

Figure 6: R_9 (saturated red rendering index) as a function of maximum achievable luminous efficacy of radiation at 2700K. Each point represents the optimal spectrum at a different CRI floor, with R_9 unconstrained. At the EU minimum (CRI $\geq$ 80), R_9 is only 21. The California JA8 minimum of $R_9 \geq 50$ is not met until CRI $\geq$ 90.

These are theoretical radiation-only efficacies assuming unit quantum efficiency. For real phosphor-converted LED lamps, Koedam and Opstelten estimated a ∼20% reduction due to realistic quantum efficiency and Stokes losses [10]. The correction is wavelength-dependent: converting a 450 nm blue pump photon to a longer-wavelength visible photon loses a fraction $1-450/\lambda_{\text{em}}$ of the photon energy as heat (the Stokes shift). Table 2 shows the penalty for representative LED phosphors.

Table 2: Stokes loss and quantum yield for LED phosphors (450 nm blue pump).

Phosphor	Emission color	λ_{em} (nm)	Stokes loss	Typ. QY	Net loss
YAG:Ce	yellow	560	19.6%	96%	23%
β-SiAlON:Eu	green	535	15.9%	83%	30%
KSF (K_2SiF_6:Mn)	red	631	28.7%	95%	33%
Nitride (CASN)	deep red	650	30.8%	88%	39%

Adding red phosphor content — needed for good R_9 — incurs a *double* penalty: (1) the Stokes loss is higher (29–31% vs. 16–20% for green/yellow), and (2) the eye's photopic sensitivity $V(\lambda)$ drops steeply in the red, so each red photon contributes fewer lumens. This is exactly why the optimizer suppresses red emission when R_9 is unconstrained, and why achieving good red rendering in real LED lamps is expensive in terms of system efficacy.

5 Discussion

5.1 The Regulatory Paradox

The results reveal a paradox in lighting regulation:

1. Regulations set a *minimum CRI* (typically 80) and a *minimum efficacy* (80–100 lm/W and rising).
2. The optimal solution for high efficacy at moderate CRI is a narrow-band, spiky spectrum.
3. Spiky spectra can achieve high CRI scores (by construction — the 8 test colors are rendered well) while having poor real-world color rendering (especially saturated colors, TCS 9–14).
4. As efficacy requirements increase over time, manufacturers are pushed further toward spiky spectra.
5. The "efficient" lamp of the future satisfies all regulations but produces light that looks *wrong* to many observers.

This is not a hypothetical concern. Commercial tri-phosphor fluorescent lamps and phosphor-converted LEDs already exploit exactly this trade-off. Their spectral power distributions, with 3–5 peaks and deep valleys, are the market's practical approximation to the mathematically optimal solution computed here.

Table 3 compares the major regulatory frameworks. The EU sets CRI ≥ 80 with no R_9 requirement; our optimizer shows that this permits $R_9 = 21$ (Table 1). The US DOE 2028 standard requires ∼125 lm/W system efficacy but imposes no color quality floor at all. Only California JA8 mandates $R_9 \geq 50$ together with CRI ≥ 90.

Table 3: Regulatory landscape for general-service lamps (2025–2026).

	EU 2019/2020	US DOE 2022	US DOE 2028	CA Title 24 JA8
Min system efficacy*	∼98	45	125	45
Min CRI (R_a)	80	—	—	90
Min R_9	—	—	—	50
CCT range	any	any	any	2700–4000 K

*lm/W for an 800 lm non-directional lamp.

5.2 Comparison with Thornton's Work

Thornton arrived at his prime-color wavelengths (450, 540, 610 nm) through a combination of psychophysical experiments and engineering intuition in the 1970s [1, 2, 3]. The `photopic` optimizer, starting from first principles (Planck's law, CIE color matching functions, von Kries adaptation), converges to the same wavelengths. This provides independent numerical confirmation of Thornton's insight.

The deeper significance is that the prime-color wavelengths are not arbitrary — they are the eigenvectors of the human visual system's color discrimination capacity, weighted by luminous sensitivity.

5.3 What the Regulations Should Do

The IES TM-30-15 standard addresses some of these problems by using 99 test color samples instead of 8, and reporting both *fidelity* and *gamut area*. Mandating minimum R_9 scores (as some California specifications do) partially closes the red-rendering loophole. But the fundamental tension between efficacy and color quality is a physical constraint that no regulation can eliminate — it can only choose where on the Pareto frontier to operate.

5.4 The Impossibility of Blackbody-Quality Light

A common assumption in lighting policy is that consumers can "buy whatever quality light they want, as long as the lamp meets efficiency standards." This assumption is false. We can show quantitatively that *no human technology enables a blackbody-quality visible spectrum to meet government efficiency standards at any warm CCT.*

5.4.1 From spectrum to wall plug

The luminous efficacy figures computed by `photopic` are *luminous efficacies of radiation* (LER): lumens per watt of *optical* power, determined solely by the shape of the spectrum. Government standards specify *system efficacy*: lumens per watt of *electrical input* at the wall plug. The two are related by a multiplicative chain of losses, each unavoidable in phosphor-converted LED technology:

Table 4: Loss chain from spectrum to wall plug in a phosphor-converted white LED lamp. "Best production" reflects top-bin parts available today; "best lab" reflects demonstrated laboratory results; "DOE limit" reflects the DOE SSL roadmap ultimate targets.

Loss factor	Best prod.	Best lab	DOE limit
Blue LED wall-plug efficiency	0.75	0.84	0.90
Phosphor backscatter/reabsorption	0.92	0.95	0.98
Thermal droop (at junction temp)	0.90	0.93	0.97
AC-DC driver efficiency	0.92	0.95	0.98
Optical (diffuser/lens/fixture)	0.90	0.95	0.97
Total	**0.514**	**0.670**	**0.813**
(total loss)	(48.6%)	(33.0%)	(18.7%)

These losses are *in addition to* the Stokes shift and quantum efficiency losses inherent in phosphor down-conversion, which are already captured by the Philips-corrected LER (Table 2). The blue LED wall-plug efficiency — the conversion of electrical power to 450 nm photons at the chip junction — is the single largest loss and the most resistant to improvement, as it is limited by Auger recombination ("efficiency droop") at practical drive currents.

5.4.2 Blackbody system efficacy

Table 5 computes the system efficacy that would be achieved by a phosphor-converted LED replicating the Planck blackbody spectrum across the visible range (380–770 nm) at several CCTs. The Philips-corrected LER accounts for Stokes and quantum-efficiency losses; the system efficacy columns further apply the loss chain from Table 4.

Table 5: System efficacy for a blackbody-spectrum phosphor-converted LED. "Vis LER" is the luminous efficacy of the visible-range blackbody radiation. "Phil LER" applies the Philips phosphor cost model. System columns multiply by the corresponding loss chain. Compare against EU Class E minimum of 85 lm/W and EU Class D minimum of 110 lm/W.

CCT (K)	Vis LER (lm/W)	Phil LER (lm/W)	Best prod. (lm/W)	Best lab (lm/W)	DOE limit (lm/W)
2100	113	68	35	45	55
2400	137	84	43	56	68
2700	156	97	50	65	79
3000	171	109	56	73	88
3500	188	123	63	82	100
4000	197	132	68	89	108
5000	203	142	73	95	116
5500	203	144	74	97	117
6500	199	146	75	97	118

The implications are stark:

- A 2700 K blackbody LED — replicating the warm white of a classic incandescent lamp — would achieve at most **65 lm/W** with today's best laboratory technology. This fails every current efficiency standard worldwide except California JA8 (which requires only 68 lm/W but also mandates CRI ≥ 90 and $R_9 \geq 50$, which a blackbody trivially satisfies).
- Even at 6500 K (harsh daylight white) with *theoretical-limit* technology that may never be achieved, the system efficacy reaches only 118 lm/W — still below the US DOE 2028 target of 120 lm/W and the forthcoming EU Class C threshold.
- At the popular residential CCTs of 2700–3000 K, a blackbody LED fails the current EU Ecodesign Class F minimum (70 lm/W) with today's production technology, and barely reaches EU Class E (85 lm/W) at 3000 K even with the most optimistic lab results.

In other words, **a consumer who wants incandescent-quality light from an LED cannot legally purchase it in the EU at any warm CCT, and this situation will not improve even with foreseeable advances in LED technology.** The efficiency regulations do not merely encourage manufacturers to optimize spectra — they make blackbody-quality illumination physically impossible to deliver at the required wall-plug efficiency.

This is the fundamental policy trade-off that spectral optimization quantifies. Our Pareto frontier results (Sections 3–4) show precisely how much color quality can be retained at each efficiency level. But the "lossless" ideal — a smooth Planckian spectrum at warm CCT — is categorically excluded by the combination of LED physics and regulatory floors. The notion that energy-efficiency regulation is "technology-neutral" is contradicted by these results: the regulations implicitly mandate non-Planckian, spectrally engineered light.

6 Literature Review

The optimization of white-light spectra has a history reaching back to the 1950s, but two papers from the 1970s are particularly relevant to the present work: Thornton's 1971 study of prime colors and Walter's 1978 application of nonlinear programming to lamp spectra.

6.1 Thornton (1971): Prime Colors and the Structure of White Light

W. A. Thornton of Westinghouse [1] posed the question: *What is the optimum spectral power distribution of artificial white light?* Starting from the CIE luminosity function $\bar{y}(\lambda)$ and the IES–CIE color rendering index, Thornton evaluated white light composed of three spectral lines. By systematic variation of each wavelength while holding the other two fixed (at 450, 540, and 610 nm), he showed that these three wavelengths are local optima of both CRI and a combined figure of merit $\mathrm{FM} = 320\eta + \mathrm{CRI}$.

Thornton's key findings:

- Three spectral lines at **450, 540, and 610 nm** — the "prime colors" — yield white light superior to commercial fluorescent lamps of the same chromaticity in *both* luminosity and CRI. Three-line light at 4200 K achieved $\eta = 0.58$, CRI = 71, compared to $\eta = 0.48$, CRI = 67 for a commercial cool-white fluorescent.

- Wavelengths near **495 and 575 nm** are "anti-prime": they degrade CRI without contributing significantly to luminosity. Removing a slice of power near 500 or 580 nm from a cool-white lamp's spectrum actually *raises* the CRI.

- A 1966 demonstration with interference filters and projectors showed that 60 observers judged three-line prime-color light as providing "very good, if not excellent" color rendering of meat, vegetables, flowers, and skin tones.

- The prime-color wavelengths correspond to the maxima of *tristimulus difference functions* $\bar{b} = \bar{z} - \bar{x} - \bar{y}$, $\bar{g} = \bar{y} - \bar{x} - \bar{z}$, $\bar{r} = \bar{x} - \bar{y} - \bar{z}$, which Thornton termed "bluminosity," "verdinosity," and "rubinosity." He proposed corresponding units (blumen, verden, ruben) analogous to the lumen, and showed that rubinosity ρ is an excellent predictor of red phosphor performance in tri-phosphor blends (Fig. 14 in the original).

Thornton's insight is that the prime-color wavelengths are not arbitrary engineering choices but reflect the structure of human color vision: they are the wavelengths that "pull hardest" toward their chromaticities per watt of power, and are therefore the most efficient at establishing a wide color gamut from a small number of spectral components.

6.2 Walter (1978): Nonlinear Programming of Lamp Spectra

Wolfgang Walter of GTE Sylvania [10] took a fundamentally different approach: rather than assuming a fixed number of spectral bands and optimizing their positions, he applied *nonlinear mathematical programming* (the generalized reduced gradient algorithm of Lasdon et al.) to find optimal spectra with *no a priori constraint on shape*. The spectrum was discretized into 31 variables (10 nm intervals from 400–700 nm), with the objective of maximizing luminous flux of a 40-watt fluorescent lamp subject to constraints on total UV excitation power, ANSI chromaticity ellipses, and a CRI floor.

Walter's key results:

- Optimal spectra with CRI up to 90–95 consist of **just four narrow bands**, positioned at approximately 460, 530, 580, and 620 nm, stable across all four ANSI lamp types (cool white, white, warm white, daylight).

- The unconstrained (CRI-free) optimum is a two-line spectrum (the MacAdam pair), confirming earlier theoretical work.

- As CRI constraints increase from 50 to 95, additional bands appear and existing bands broaden, with a monotonic decrease in luminous efficacy (from ∼140 lm/W at CRI 50 to ∼107 lm/W at CRI 95 for cool white).
- The optimization surface is "fairly flat" near the optimum: small perturbations to the SPD do not greatly affect luminance or CRI, suggesting practical tolerance for real phosphor emission profiles.
- When *all* component indices $R_j \geq C$ are required (not just the average R_a), efficacy drops further — confirming that the CRI average conceals weaknesses in individual color samples.

The published discussion section is notable: Koedam and Opstelten of Philips argued that the four-band optimum offers "no real improvement" over the Philips three-band lamp after correcting for realistic quantum efficiency (∼80% rather than 100%). Thornton himself commented, noting that if subjective criteria (color preference, visual clarity, perceived brightness) are considered rather than CRI alone, three bands may be superior to four.

6.3 Relationship to Our Results

Our `photopic` optimizer, approaching the problem from yet another direction — continuous piecewise-linear log-amplitude perturbation of a blackbody, optimized by COBYLA — independently converges to essentially the same spectral structure. The three-peak topology that dominates our results at low CRI floors (437 lm/W at CRI $\geq$ 60) corresponds to Thornton's prime colors, while the broadening and addition of a fourth peak at higher CRI floors (CRI $\geq$ 90 and above) matches Walter's four-band result. The wavelength positions we find (∼450, ∼540, ∼610 nm) agree with Thornton to within the resolution of our spectral parameterization.

The consistency across three independent methods — Thornton's parametric three-line analysis, Walter's unconstrained nonlinear programming, and our hierarchical COBYLA optimization — provides strong evidence that the prime-color structure is a genuine feature of the CRI/efficacy optimization landscape, not an artifact of any particular numerical method.

7 The Deficiencies of CRI and the Legislative Paradox

7.1 Why CRI is Known to be Wrong

The CIE Color Rendering Index (CRI, R_a) has been recognized as flawed since at least the 1980s, yet it remains the dominant regulatory metric. The principal deficiencies are:

1. **Only 8 unsaturated test samples.** R_a averages R_1 through R_8, all of which are low-chroma, pastel-like Munsell samples. Saturated colors — which are the ones most affected by narrow-band spectra — are represented only in R_9–R_{14}, which are reported but *not* included in the official score. This is the loophole our optimizer exploits: it can drive R_9 (saturated red) to -28 while maintaining $R_a = 65$.
2. **Von Kries adaptation model.** The CRI uses a simple diagonal (von Kries) chromatic adaptation transform, which is known to be a poor approximation of human color constancy, especially for narrow-band illuminants far from the blackbody locus.
3. **Uniform color space.** The CRI computes color shifts in the CIE 1964 $U^*V^*W^*$ space, which is perceptually nonuniform. Equal ΔE values do not correspond to equal perceived color differences.

4. **Averaging hides problems.** A lamp with seven $R_i = 95$ and one $R_i = 25$ gets $R_a = 86$ — an apparently acceptable score that conceals a severe deficiency in one color region. Walter's work showed that requiring all $R_j \geq C$ rather than just $R_a \geq C$ significantly changes the optimal spectrum and reduces achievable efficacy.

5. **Penalizes increased saturation.** Because CRI measures *fidelity* (closeness to the reference illuminant), a lamp that renders colors *more* saturated than the reference — which people often prefer — receives a *lower* CRI score. Thornton's color preference index (CPI) and the IES TM-30-15 standard [14] address this.

7.2 Why Legislators Use It Anyway

Despite these well-known deficiencies, CRI remains the basis for lighting regulations worldwide (EU Ecodesign 2019/2020, US DOE/EISA, California Title 24, Energy Star). The reasons are partly institutional inertia and partly the absence of a universally agreed replacement:

- **IES TM-30-15/18** defines two metrics: R_f (fidelity, similar to CRI but with 99 test samples and CAM02-UCS) and R_g (gamut area, measuring saturation changes). It is technically superior but has not yet been mandated by any major regulatory body.
- **CIE 224:2017** proposed a general color fidelity index R_f similar to TM-30, but adoption has been slow.
- The lighting industry has decades of CRI-referenced product data, specifications, and marketing materials. Changing the metric would require re-characterizing the entire installed base.

The result is a regulatory landscape that *incentivizes* the very spiky, CRI-gaming spectra that our optimizer finds. Manufacturers who maximize efficacy while just meeting CRI ≥ 80 are rewarded by the regulatory framework, even though the resulting light may render saturated colors poorly. The "weird final state" is not a theoretical curiosity — it is the market equilibrium toward which regulations push.

8 Philips Phosphor Cost Model and R_9 Maximization Sweeps

The luminous efficacy of radiation (LER) computed in Sections 3–4 assumes unit quantum efficiency: every watt of optical power in the spectral power distribution produces its full complement of lumens. In a real phosphor-converted LED, shorter-wavelength photons from the blue pump must be down-converted to longer wavelengths by phosphors, incurring a Stokes loss proportional to the wavelength ratio. A photon at 450 nm converted to 610 nm loses $1 - 450/610 \approx 26\%$ of its energy as heat, regardless of quantum yield.

Koedam and Opstelten of Philips [10] proposed a simple cost model that captures this loss: the cost of producing one watt of optical power at wavelength λ relative to the pump wavelength λ_p is

$$C(\lambda) = \max\left(1,\ \frac{\lambda}{\lambda_p \cdot \eta_Q}\right) \tag{4}$$

where η_Q is the phosphor quantum yield (here assumed 0.95). Photons at the pump wavelength cost 1 (no conversion needed); longer wavelengths cost progressively more due to the Stokes shift. The

"Philips-corrected LER" replaces the radiant power denominator with the cost-weighted integral:

$$\eta_{\text{Phil}} = K_m \frac{\int S(\lambda)\, V(\lambda)\, d\lambda}{\int S(\lambda)\, C(\lambda)\, d\lambda} \tag{5}$$

We use this corrected efficacy as the constraint in a new optimization mode: **maximize** R_9 (saturated red rendering) subject to $\text{CRI}(R_a) \geq \text{floor}$, $\text{CCT} \in [T_0 - 50, T_0 + 50]$, $D_{uv} \leq 0.012$, and $\eta_{\text{Phil}} \geq E_{\min}$. The CRI constraint uses a double-sided "pin" ($R_a \in [\text{floor}, \text{floor} + 2]$) to prevent the optimizer from spending efficacy budget on CRI headroom rather than R_9.

8.1 CRI $\geq$ 80 Sweep

Table 6 shows the results of sweeping the Philips efficacy floor from 200 to 320 lm/W at 2700 K with CRI pinned to 80 ± 2.

Table 6: Maximum R_9 vs. Philips-corrected efficacy floor at 2700 K, CRI $\geq$ 80. All results at 129 spectral parameters. LER is the raw luminous efficacy of radiation.

$E_{\min}$ (Phil)	CRI	R_9	LER	CCT	D_{uv}
200	81.1	99.2	297	2663	0.010
210	81.8	97.3	309	2670	0.011
220	81.4	98.5	323	2703	0.011
230	82.5	98.5	335	2701	0.011
240	80.0	97.8	349	2700	0.010
250	80.0	96.1	362	2691	0.012
255	82.2	89.7	364	2750	0.012
260	82.0	89.0	370	2750	0.012
265	82.2	88.2	377	2750	0.012
270	82.1	47.1	385	2650	0.012
275	81.9	40.2	391	2650	0.013
280	81.2	44.7	397	2650	0.015
285	81.6	35.0	404	2648	0.015
290	80.2	32.7	410	2646	0.016
295	76.6	29.1	415	2644	0.018
300	76.1	25.2	425	2646	0.016
305	72.3	29.7	428	2651	0.020
310	69.9	25.4	431	2724	0.022
315	67.9	23.6	435	2762	0.023
320	66.2	16.3	439	2762	0.023

The results reveal three distinct regimes, separated by horizontal rules in the table:

1. **Comfortable region** ($E_{\min} \leq 265$): $R_9 > 88$, CRI $\approx$ 80–82. The efficacy budget is generous enough for the optimizer to produce spectra with excellent red rendering. The spectrum remains relatively smooth with broad red content.

2. R_9 **cliff** ($E_{\min}$ = 270–290): R_9 collapses from 88 to 33. The optimizer can still satisfy CRI $\geq$ 80 (since R_9 is not part of R_a), but cannot afford the red phosphor power needed for saturated red fidelity. This is the regime where the CRI loophole is most actively exploited.

3. **Constraint infeasible** ($E_{\min} \geq 295$): Even CRI = 80 becomes unachievable. The optimizer violates the CRI constraint (dropping to 66–77) and the chromaticity tolerance (D_{uv} exceeds 0.012). At these extreme efficacies, no well-rendering white light is possible.

8.2 CRI ≥ 95 Sweep

Table 7 shows the same sweep at CRI ≥ 95. Higher CRI forces broader spectra, which leaves less room for the R_9 loophole — but also reduces the achievable efficacy range.

Table 7: Maximum R_9 vs. Philips-corrected efficacy floor at 2700 K, CRI ≥ 95. All results at 129 spectral parameters.

$E_{\min}$ (Phil)	CRI	R_9	LER	CCT	D_{uv}
200	96.3	99.4	290	2776	0.002
210	97.0	98.2	304	2746	0.002
220	97.0	97.5	318	2745	0.003
230	95.0	97.1	331	2750	0.005
240	95.0	96.5	344	2750	0.005
250	95.1	94.9	358	2750	0.007
255	95.0	85.7	365	2663	0.005
260	94.0	92.4	370	2751	0.009
265	94.7	92.5	377	2750	0.006
270	90.5	85.0	380	2706	0.010
275	87.9	83.1	385	2700	0.013
280	93.3	58.8	396	2752	0.008
285	85.8	76.5	396	2691	0.015
290	84.2	61.7	400	2679	0.014
295	85.2	66.7	408	2748	0.016
300	79.6	54.0	410	2657	0.018
305	79.5	52.8	416	2681	0.018

Compared to the CRI ≥ 80 sweep, the CRI ≥ 95 results show:

- The R_9 cliff occurs at a similar efficacy (∼270 Phil lm/W) but is less severe: R_9 drops from 92 to 85 rather than from 88 to 47. The broader spectra required by CRI ≥ 95 provide some red content "for free" as a byproduct of the fidelity constraint.
- Constraint infeasibility sets in earlier: by $E_{\min} = 280$, the optimizer can no longer reliably achieve CRI = 95 (dropping to 84–93). The nonconvex optimization landscape also produces non-monotonic behavior (e.g., $R_9 = 58.8$ at 280 but 76.5 at 285), reflecting different local optima.
- The feasible range is narrower: useful results span 200–265 Phil lm/W for CRI ≥ 95 vs. 200–290 for CRI ≥ 80.

8.3 The R_9 Cliff: Policy Implications

Both sweeps confirm the central finding of this report: there exists a critical Philips-corrected efficacy threshold (∼265–270 lm/W at 2700 K) above which saturated red rendering degrades sharply.

Applying the production loss chain from Table 4 (best production: $\times 0.514$), this corresponds to a system efficacy of $\sim$135–140 lm/W — squarely in the range of current and near-future regulatory floors.

Regulators who set only a CRI ≥ 80 floor are implicitly permitting R_9 values as low as 33–47 (Table 6), representing severely compromised red rendering. Even CRI ≥ 95 permits R_9 to drop below 60 at high efficacy. Only explicit R_9 mandates — as in California JA8 — can close this loophole.

9 Grid-Based Integration Optimization

9.1 Motivation

Each evaluation of the COBYLA objective function requires computing CRI (14 color shifts), CCT (root-finding on the blackbody locus), LER, and chromaticity — a total of approximately **99 Romberg integrations** per evaluation. The breakdown:

- CRI calculation: 14 TCS $\times$ 3 tristimulus values (X, Y, Z) = 42 integrations for TCS color coordinates, plus 3 for the test illuminant's own XYZ, plus 3 for the reference blackbody = 48 integrations
- Chromaticity and LER: 4 integrations (X, Y, Z, total luminous flux)
- Philips cost model: 1 integration (cost-weighted power)
- CCT search: multiple iterations of the above during Brent's method root-finding

Each Romberg integration (Numerical Recipes §4.4, midpoint rule variant) uses adaptive refinement, typically evaluating the integrand at 50–200 wavelength points. At 129 spectral parameters, COBYLA makes thousands of objective evaluations over the 8-level hierarchical run, resulting in optimization times of $\sim$50 minutes per sweep point.

9.2 Approach: Pre-Tabulated Weight Vectors

The key observation is that all integrands have the form $S(\lambda) \cdot W(\lambda)$ where $S(\lambda)$ is the test spectrum (which changes every evaluation) and $W(\lambda)$ is a fixed weight function (CIE color matching functions, $V(\lambda)$, TCS reflectances, Philips cost, etc.) that depends only on the problem setup.

By discretizing the visible range onto a fixed grid — 391 points at 1 nm spacing from 380 to 770 nm — we can:

1. Pre-tabulate all weight vectors $\mathbf{w}_k$ at grid points once during initialization
2. Sample the test spectrum $\mathbf{s}$ onto the grid (one pass through the M3 `ParametricSpectrum.Eval` function)
3. Replace each integration $\int S(\lambda)W(\lambda)d\lambda$ with a dot product $\Delta\lambda \cdot \mathbf{s}^T \mathbf{w}_k$

The pre-tabulated weight vectors include:

- CIE $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$ — 3 vectors
- $K_m V(\lambda)$ (photopic conversion) — 1 vector

- $C(\lambda)$ (Philips cost) — 1 vector
- 14 TCS reflectances $\times\ \bar{x}, \bar{y}, \bar{z}$ — 42 pre-multiplied vectors of the form $\mathrm{TCS}_j(\lambda) \cdot \bar{x}(\lambda)$

Each "integration" becomes a vector dot product in $\mathbb{R}^{391}$ — a tight loop of 391 multiply-accumulates with no function calls, no adaptive refinement overhead, and no M3/Scheme boundary crossings beyond the initial spectrum sampling.

9.3 Pure-Scheme Blackbody

The reference illuminant (blackbody at the CCT) must also be computed on the grid. Rather than making 391 calls to the M3 `Blackbody.PlanckRadiance` function, we implemented Planck's law directly in Scheme:

$$B_\lambda(T, \lambda) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\exp(hc/\lambda k_B T) - 1} \tag{6}$$

where h, c, and k_B are defined as Scheme constants. This eliminates the M3/Scheme call overhead for the reference illuminant and keeps the entire grid-mode calculation in pure Scheme except for the initial spectrum sampling.

9.4 Implementation

The grid mode is activated by the `-grid` command-line flag and can be toggled at runtime via the `*use-grid*` variable. This allows direct comparison of results between the Romberg and grid methods on the same optimization problem.

A critical implementation detail is the normalization convention: the CIE CRI calculation normalizes spectra such that $X + Y + Z = 100$, not $Y = 100$. The `photopic` codebase follows the convention that `xyz->Yxy` returns $(X + Y + Z, x, y)$ as its first element (confusingly named "Y" in the function name but actually the sum). Getting this wrong in the grid normalization caused CRI errors of up to 63 units in R_9 during development — a subtle but critical bug.

9.5 Accuracy Validation

Table 8 compares the grid and Romberg methods on the same optimized spectrum (CRI ≥ 80, Philips eff ≥ 300, 129 parameters).

Table 8: Grid vs. Romberg accuracy on an optimized 129-parameter spectrum.

Metric	Romberg	Grid (1 nm)	Difference
CRI (R_a)	81.93	81.76	0.18
R_9	40.15	39.53	0.62
Phil LPW	274.31	274.24	0.07
LER	424.0	423.8	0.2

The differences are negligible for optimization purposes. Testing at 0.5 nm grid spacing (781 points) produced identical values to 1 nm, confirming that the grid is converged.

9.6 Performance Results

Table 9: Timing comparison: Romberg vs. grid integration.

Measurement	Romberg	Grid (1 nm)
Per objective evaluation	0.59 s	0.20 s
Full 8-level optimization	52 min	14.5 min
Speedup	**3.6**×	

The 3.6× speedup makes parameter sweeps practical: a 20-point sweep that took 17 hours with Romberg integration completes in under 5 hours with the grid method. The sweep results in Tables 6 and 7 were computed using grid-mode integration.

Because the grid and Romberg methods produce slightly different objective landscapes (due to <0.2% numerical differences), the COBYLA optimizer may converge to different local optima. This is expected behavior in nonconvex optimization and is not a deficiency of the grid method — the solutions are equally valid.

10 TM-30: A Better Color Metric

10.1 Why CRI Needs Replacement

The deficiencies of CRI enumerated in Section 7 are not merely academic: our optimization results demonstrate that they have practical consequences. The CRI loophole (R_9 not included in R_a) allows the optimizer to produce spectra with $R_9 = -28$ while claiming CRI = 65. The 8-sample average hides severe rendering failures. The von Kries adaptation model and CIE 1964 $U^*V^*W^*$ color space are outdated by decades of color science research.

The IES TM-30-18 standard [14] was developed specifically to address these problems. It is not merely an incremental improvement — it represents a fundamentally different approach to characterizing color rendition.

10.2 TM-30 Architecture

TM-30 evaluates light sources using **99 Color Evaluation Samples** (CES), selected by Royer et al. from a database of 105,000 measured object spectra to span the full range of real-world colors including saturated hues [16]. Compare this to CRI's 8 unsaturated Munsell samples (or 14 including the supplementary set).

The colorimetric pipeline replaces every component of the CRI calculation with modern equivalents:

Component	**CRI (1995)**	**TM-30 (2018)**
Test samples	8 (unsaturated)	99 (full gamut)
Color space	CIE 1964 $U^*V^*W^*$	CAM02-UCS (J', a', b')
Adaptation	von Kries diagonal	CIECAM02 (CAT02)
Scale	Linear ($R_i = 100 - 4.6\Delta E$)	Logarithmic
Metrics	R_a (fidelity avg)	R_f, R_g, $R_{cs,hj}$

10.2.1 CIECAM02 Color Appearance Model

TM-30 computes all color differences in the CIECAM02 color appearance model [15], which accounts for:

- **Chromatic adaptation**: The CAT02 transform (a 3×3 matrix applied in an LMS-like cone response space) models how the human visual system adapts to the illuminant white point. This is substantially more accurate than von Kries scaling, especially for narrow-band illuminants.
- **Nonlinear response compression**: Cone signals are compressed by a power function modeling the saturation of neural response, parameterized by the adapting luminance.
- **Perceptual correlates**: The model outputs lightness (J), chroma (C), hue angle (h), colorfulness (M), and saturation (s) as perceptual dimensions rather than tristimulus coordinates.

For TM-30, the CIECAM02 J, a, b correlates are further transformed into the CAM02-UCS (Uniform Colour Space) via:

$$J' = \frac{1.7J}{1 + 0.007J} \tag{7}$$

$$M' = \frac{1}{0.0228} \ln(1 + 0.0228M) \tag{8}$$

$$a' = M' \cos h, \qquad b' = M' \sin h \tag{9}$$

This logarithmic compression yields a more perceptually uniform space than the CIE $U^*V^*W^*$ used by CRI.

10.2.2 TM-30 Metrics

TM-30 reports three families of metrics:

1. **Fidelity Index** R_f: Analogous to CRI R_a, but averaged over all 99 CES in CAM02-UCS. The color difference $\Delta E'$ for each sample is mapped through a logarithmic function:

$$R_{f,j} = 10 \ln\left(\exp\left(\frac{100 - c_f \Delta E'_j}{10}\right) + 1\right) \tag{10}$$

   where c_f is a normalization constant. This ensures $R_f \leq 100$ (equality only for a perfect match) and provides better discrimination at high fidelity than CRI's linear scale. $R_f = \frac{1}{99} \sum_j R_{f,j}$.

2. **Gamut Index** R_g: Measures whether the test source expands or contracts the color gamut relative to the reference. The 99 CES are assigned to 16 hue bins; R_g is the ratio of the gamut areas (in $a'b'$ coordinates) of the test and reference illuminants, scaled so that $R_g = 100$ means identical gamut area. Values > 100 indicate gamut expansion (more saturated colors); < 100 indicates gamut contraction. CRI penalizes both; R_g distinguishes them.

3. **Local chroma shift** $R_{cs,hj}$: For each of the 16 hue bins, TM-30 reports whether colors in that hue region become more or less saturated. $R_{cs,h1}$ (the first hue bin, centered on red) is of particular interest: it directly measures the red chroma shift that R_9 only crudely captures. Negative $R_{cs,h1}$ means reds are desaturated — the precise failure mode our optimizer exploits.

10.3 Why TM-30 Matters for This Work

The TM-30 metrics would fundamentally change the optimization landscape:

- **No 8-sample loophole.** With 99 CES spanning the full gamut, the optimizer cannot sacrifice one color region to boost another. Saturated reds are represented by multiple CES samples, not just the single TCS 9.
- $R_{cs,h1}$ **as a direct red metric.** Rather than the blunt instrument of R_9 (which conflates hue shift, chroma shift, and lightness shift into a single number), $R_{cs,h1}$ isolates the red chroma shift. This is precisely the quantity that determines whether reds "look right."
- **Better adaptation model.** The CIECAM02 transform handles the narrow-band illuminants produced by our optimizer more accurately than von Kries scaling. CRI may over- or under-penalize color shifts for these non-Planckian spectra.
- **Gamut shape information.** R_g and the per-hue-bin metrics reveal *how* colors change, not just *how much*. A spectrum that expands the blue gamut while contracting red would show $R_g \approx 100$ (gamut area preserved) but $R_{cs,h1} < 0$ and $R_{cs,h9} > 0$ — information that CRI R_a completely hides.

10.4 Implementation

TM-30 is implemented in pure Scheme within `photopic.scm`, requiring no changes to the Modula-3 binary. The implementation comprises approximately 370 lines of Scheme code plus tabular data:

1. **99 CES reflectance spectra and CIE 1964** 10° **CMFs** (ces99.scm, ∼4300 lines): Extracted from the `colour-science` library [18] (which provides the CIE 2017 dataset) and tabulated at 1 nm resolution. The file also contains the CIE D-illuminant basis functions $S_0(\lambda)$, $S_1(\lambda)$, $S_2(\lambda)$ for reference illuminant construction.
2. **CIECAM02 forward model** (∼90 lines): Full forward transform from XYZ to (J, M, h), including CAT02 chromatic adaptation, Hunt–Pointer–Estévez cone response space, nonlinear compression, opponent channels, eccentricity factor, and the complete lightness/chroma computation. Viewing conditions use average surround ($c = 0.69$, $N_c = 1$, $F = 1$) with $L_A = 100\,\mathrm{cd/m^2}$ and $Y_b = 20$.
3. **CAM02-UCS transform** (∼8 lines): The $(J, M, h) \to (J', a', b')$ mapping with $K_L = 1.0$, $c_1 = 0.007$, $c_2 = 0.0228$.
4. **Reference illuminant selection** (∼50 lines): Following CIE 224:2017, the reference is a Planckian blackbody for CCT $< 4000\,\mathrm{K}$, a CIE D-series illuminant for CCT $> 5000\,\mathrm{K}$, and a luminance-normalized blend for the 4000–5000 K transition zone. The D-illuminant is constructed from CCT via the standard daylight-locus chromaticity polynomials and the $S_D(\lambda) = S_0 + M_1 S_1 + M_2 S_2$ basis decomposition.
5. **Metric computation** (∼80 lines): For each of the 99 CES, reflected XYZ values (via pre-tabulated weight-vector dot products, matching the grid pattern used for CRI) are transformed through CIECAM02 and CAM02-UCS, color differences are computed, and samples are assigned to 16 hue bins. R_f uses the logarithmic soft-cap formula, R_g uses the shoelace polygon area formula, and R_{cs,h_j} is the per-bin chroma shift.

Validation. The implementation has been validated against the `colour-science` Python library (v0.4.7) on multiple illuminants: Planckian 3000 K ($R_f \approx 100$, $R_g \approx 100$ as expected for self-reference), FL2 ($R_f = 70.3$ vs. reference 70.1, $R_g = 86.4$ vs. 86.4), and FL11 ($R_f = 80.1$ vs. 80.0, $R_g = 99.9$ vs. 101.1). All results agree within ± 2 units. The CRI computation is unchanged (regression-verified).

Performance. TM-30 adds approximately 30% overhead to each grid-mode spectrum evaluation, consistent with the pre-implementation estimate. The 99 CES weight vectors are pre-tabulated at startup alongside the existing 14 TCS vectors.

11 Future Work

Several directions for further work remain, building on the results reported above:

1. **TM-30 optimization.** With the TM-30 implementation now complete (Section 10), the next step is to use R_f and R_g as optimizer constraints, replacing or supplementing the CRI constraints. Re-running the Pareto sweeps with $R_f \geq 80$ instead of $R_a \geq 80$ would reveal how much of the CRI loophole exploitation survives under a modern color metric. The R_{cs,h_j} per-hue-bin metrics could also be used as constraints to prevent selective desaturation of specific hue regions.

2. **Multiple CCTs.** Extend the sweeps of Section 8 to the full range of commercial color temperatures (2700–6500 K). Walter showed that the four-band positions are remarkably stable across ANSI lamp types; verifying this with our continuous parameterization and Philips cost model would be valuable.

3. **LED-realistic spectra.** Constrain the spectrum to physically realizable LED spectra: a narrow blue pump peak ($\sim$450 nm, 20 nm FWHM) plus one or more phosphor emission bands (50–100 nm FWHM, constrained shapes). This would bridge the gap between our theoretical optima and practical LED design.

4. **Scotopic/mesopic efficacy.** Outdoor and roadway lighting operates at mesopic light levels where the scotopic sensitivity function $V'(\lambda)$ contributes. The S/P ratio (scotopic to photopic luminance ratio) is an additional design variable that regulations are beginning to address.

5. **Circadian impact.** Blue-rich spectra suppress melatonin production. A constraint on melanopic equivalent daylight illuminance (M-EDI) could be added to the optimization, capturing the tension between visual efficacy and biological impact.

12 Conclusion

The constrained optimization of light spectra reveals that regulatory pressure on luminous efficacy drives lamp spectra toward a mathematically determined "weird" final state: three narrow spectral peaks near 450, 540, and 610 nm with suppressed emission elsewhere. This state maximizes lumens per watt while technically satisfying color rendering requirements, but at the cost of poor rendering of saturated colors — especially reds.

The `photopic` tool, built as a hybrid Modula-3/Scheme application, enables rapid exploration of this trade-off space using COBYLA constrained optimization with hierarchical spectral resolution.

Availability

The `photopic` tool, this report, the optimization data, and all plotting scripts are open source. The source code is available in two repositories:

- **Photopic tool and data:** `https://github.com/IntelLabs/async-toolkit`, branch `claude_edits`, path `m3utils/m3utils/photopic/`. The report and optimization results are in `photopic/report-runs/`.
- **CM3 compiler, MScheme, and runtime libraries:** `https://github.com/modula3/cm3`. MScheme lives in `m3-scheme/`; the COBYLA binding, spectral libraries, and CIE data are in the `m3utils` tree above.

Both repositories are licensed under open-source terms (Apache 2.0 for the async-toolkit; BSD for CM3).

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